

In Memoriam: C.A.R. Hoare

Knighthood Turing Laureate Tony Hoare developed the Quicksort algorithm, Hoare Logic, and provided many other fundamental contributions underlying modern computing.

TURING AWARD WINNER Sir Charles Antony Richard Hoare died March 5, 2026, at the age of 92. His 1980 ACM A.M. Turing Award cited “fundamental contributions to the definition and design of programming languages,” but that is only part of what he contributed during a career spanning six decades. Tony, as he was universally known, was feted for his creation of Quicksort, the development of Hoare Logic, the introduction of the monitor concept for operating systems, and many other fundamental contributions underlying modern computing.

Hoare was born on Jan. 11, 1934, in Colombo, Ceylon (now Sri Lanka), where his father was a member of the English Crown’s colonial administration. In school, everyone called him “professor” because he was studious and uncommunicative, he recalled in a 2009 interview.⁶ His favorite authors were Bernard Shaw and Bertrand Russell. He was educated in England at the Dragon School in Oxford and the King’s School in Canterbury; he entered Merton College, Oxford, in 1952 to read classics and philosophy.

After graduating from Merton, Hoare spent 18 months studying Russian as part of his compulsory National Service in the Royal Navy. He returned to Oxford in 1958 for a postgraduate certificate in statistics, where he encountered computer programming for the first time, learning Autocode on a Ferranti Mercury. In 1959, he traveled to Moscow State University as a British Council exchange student to study under Andrey Nikolayevich Kolmogorov. There, he was tasked with studying automatic translation between Russian and English. “I came to the conclusion that it wasn’t going to be possible.”

Before that realization, Hoare developed Quicksort, which he later described as “the only really interesting algorithm I ever developed.”¹ He origi-

nally designed it to sort the words of Russian sentences in memory before looking them up on magnetic tape, which minimized tape movement during access.

Hoare went to an exhibition in Moscow where Elliott Brothers, Ltd., an early English computer company, was showing one of its small scientific systems. “Because of my interest in computers, I spent quite a bit of time at their stand,” Hoare recalled. The salespeople gave him a ride back to his apartment and offered him a job the next day, “with an additional 100 pounds a year on my salary because I knew Russian.”⁶

Back in England, Hoare started working on another language translator: a compiler that transformed source code in the Algol 60 programming language into machine code for the Elliott 803 computer—a transistorized, bit-serial machine with a 39-bit word. “My first task was to implement for the new Elliott 803 computer, a library subroutine for a new fast method of internal sorting just invented by Shell,” Hoare wrote in 1981. But after he implemented Shell’s algorithm, he told his boss he had invented a method that usually ran faster than Shell Sort. “He bet me sixpence that I had

not,” Hoare wrote. In fact, he had, and he won the bet and published the algorithm the following year.^{2,3}

It was while working on the compiler project that Hoare met Jill Pym, a programmer a few months older than him and, he recalled, far superior; she had been working at Elliott Brothers since 1956. They married on Jan. 13, 1962 and had three children. She left the company in 1962 but continued to be an active computer scientist, working on scientific computing and the ALGOL language and eventually as a system programmer on the British National Health Service “Fourth Dimensions” admissions system.

It was also during this period that Hoare and Niklaus Wirth designed ALGOL W, a proposed successor to ALGOL 60. He introduced null references in its type system—a feature that seemed expedient at the time. Decades later, at a software conference in 2009, he offered a now-famous mea culpa: “I call it my billion-dollar mistake. At that time, I was designing ... the first comprehensive type system for references in an object-oriented language. I couldn’t resist the temptation to put in a null reference, simply because it was so easy to implement. This has led to innumerable errors, vulnerabilities, and system crashes, which have probably caused a billion dollars of pain and damage in the last forty years.”

Promoted to chief engineer at Elliott, Hoare led a larger operating system project that ultimately failed. Reading about IBM’s OS/360 afterward, he recalled thinking, “Wow, these people must be very clever, how could they do it and [not me]?”—only to discover later “that they couldn’t either.” When Elliott merged with another firm in 1968 and canceled his projects, Hoare secured the Professorship of Computing Science at Queen’s University in Belfast, with the goal of developing approaches to make reliable concurrent software easier to write.

He expressed a deep conviction that the principles of logic and clear expression mattered as much in software as in any other intellectual endeavor.



The only approach that would work, he realized, was one that used the unforgiving strictures of formal symbolic mathematics to analyze the mess software had become. His landmark 1969 paper, “An Axiomatic Basis for Computer Programming,”⁴ published in *Communications*, introduced what became known as Hoare logic—a formal system for reasoning about the correctness of computer programs using preconditions and postconditions. ACM Fellow Moshe

Vardi observes, “[That paper] lifted a flag that had a large following for over 50 years now. But he was a gentle giant, always polite and always low-key. When I expressed, a few years ago, a slight disagreement with one aspect of the 1969 paper, he replied politely and constructively. But his dream of verified programs is alive and well, and getting a new push from recent progress in proof assistants and large language models.”

Formal methods soon found their

professional home in Working Group 2.3 on Programming Methodology of the International Federation of Information Processing (IFIP). Although Hoare was not a founding member, he attended the first meeting and is widely credited with helping to make it a friendly, mutually supportive research community. Hoare used the group as an incubator for several foundational concepts in computer science, including what is now known as the Hoare Triple and methods for

scaling mathematical proofs of correctness to cover complex programming constructs, such as iteration, local variables, procedures, and jumps. His work heavily influenced other WG 2.3 members, including Edsger W. Dijkstra and David Gries, shaping the group's overall shift toward formal verification.

Hoare moved to the University of Oxford in 1977, where he led the Programming Research Group and established new courses in computer science, including in software engineering. The following year, he made another groundbreaking contribution with "Communicating Sequential Processes," also published in *Communications*.⁵ In it, he proposed a formal language for describing patterns of interaction between concurrent programs through pre-planned communications. The paper became one of the most cited in computer science, and the ideas were elaborated into a full theory in his 1985 book of the same title. CSP has been hugely influential in both software design and hardware architecture.

"Computer Science was tiny when he joined Oxford," recalls Bill Roscoe, who received his D.Phil. from Oxford in 1982 with Hoare as his thesis advisor. Roscoe remained at Oxford, eventually serving as department head from 2003 to 2014. He said of Hoare, "He developed it firstly into a small department in his own areas and then broadened it out into a much bigger group, though one much smaller than it is today." Roscoe noted, "He certainly knew how to get the best out of me and his other protégés. He was always extremely loyal to these people."

ACM Fellow and Microsoft Distinguished Scientist Sriram Rajamani recalls an email he received from Hoare after he and his collaborator sent Hoare a draft of a paper they were working on. "Rather than offering generic praise, he explained precisely why the ideas mattered," citing several specifics. "That email set a lasting standard for me for how senior researchers can engage seriously with the work of others. Tony's email also pointed out some limitations, and he offered to collaborate with us to do further work and remedy the limitations."

Among Hoare's other notable con-

PROFESSOR MOSHE VARDI

Tony Hoare was one of the founding fathers of computer science.

tributions were the monitor concept for structuring concurrent access to shared resources in operating systems: a higher-level, less error-prone alternative to semaphores that became a standard construct in languages such as Java. He also originated the case statement construct used in virtually every modern programming language.

In 1999, Hoare retired from Oxford at the mandatory age of 65. He then joined Microsoft Research in Cambridge, England, fulltime, having served as a consultant there since 1997. He held the title of Senior Principal Researcher until 2015 while working on theories, tools, and experimental methods for producing verified, reliable software. He continued as an honorary visitor until 2021, when he was appointed Emeritus Researcher.

Hoare received numerous recognitions and accolades for his many contributions. After becoming the seventeenth Turing Award recipient in 1980, he was awarded the Harry H. Goode Memorial Award in 1981, elected as a Fellow of the Royal Society in 1982, received the Faraday Medal in 1985, and the IEEE Computer Pioneer Award in 1990. In 2000, he was awarded both the Kyoto Prize in Advanced Technology (Information Science) and a knighthood from Queen Elizabeth II. He received the IEEE John von Neumann Medal in 2011, the Royal Medal (Physical) from the Royal Society in 2023, and was inducted as a Fellow of the Computer History Museum in 2006 and of the ACM in 2020. He held 15 honorary doctorates and was a foreign member of both the U.S. National Academy of Sciences and the National Academy of Engineering, as well as a Fellow of the Royal Academy of Engineering.

Throughout his career, he ex-

pressed a deep conviction that the principles of logic and clear expression mattered as much in software as in any other intellectual endeavor. "The price of reliability is the pursuit of the utmost simplicity. It is a price which the very rich find most hard to pay," he once observed.

Rajamani recalls, "What struck me most about Tony was not his stature—though everyone knew it—but how little he relied on it. In our interactions, Tony was first and foremost a colleague. He engaged deeply with ideas, asked precise questions, and cared intensely about clarity—especially about what it really means for a system to be correct. Working with him set a very high bar, but it never felt intimidating. His humility made rigor feel accessible." He further recalled, "What stayed with me just as much were the personal gestures. Tony and his wife Jill invited us to their home for supper before the workshop and made us feel not like visitors, but like colleagues. Those moments mattered: They conveyed that intellectual seriousness and human warmth were not in tension."

As Vardi summarized: "Tony Hoare was one of the founding figures of computer science."

Hoare is survived by his wife, Lady Hoare, and two children, Thomas Hoare and Joanna Francis; his youngest son, Matthew, predeceased him. ■

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